

Module - 2

Scheduling Problems:

FCFS Scheduling

- ① Consider the following set of processes that arrive at the following time shown in table with burst-time. Calculate Average waiting time and Average turnaround time by applying FCFS

Pid	Burst time
P ₁	24
P ₂	3
P ₃	3

assume arrival time of all processes are 0.

②

Pid	AT	BT
P ₁	0	2
P ₂	3	1
P ₃	5	6

③

Pid	AT	BT
P ₁	0	8
P ₂	1	4
P ₃	2	9
P ₄	3	5

④

Pid	AT	BT
P ₁	0	4
P ₂	1	3
P ₃	2	1
P ₄	3	2
P ₅	4	5

⑤

Pid	AT	BT
P ₁	2	3
P ₂	1	6
P ₃	2	4
P ₄	0	8
P ₅	1	2

⑥

Pid	AT	BT
P ₁	0	3
P ₂	1	1
P ₃	2	1
P ₄	3	2

⑦

Pid	AT	BT
P ₀	0	7
P ₁	1	3
P ₂	2	7

⑧

Pid	AT	BT
P ₁	0	5
P ₂	1	7
P ₃	3	4

Consider the following set of processes with length of CPU burst given in millisecond units, find out Average waiting time and Average turnaround time by applying SJF - non-preemptive.

(3)

Pid	AT	BT
P ₁	0	8
P ₂	1	4
P ₃	2	9
P ₄	3	5

(4)

Pid	AT	BT
P ₁	0	12
P ₂	2	4
P ₃	3	6
P ₄	8	5

(5)

Pid	AT	BT
P ₁	1	7
P ₂	2	5
P ₃	3	1
P ₄	4	2
P ₅	5	8

(6)

Pid	AT	BT
P ₁	0	7
P ₂	1	5
P ₃	2	3
P ₄	3	1
P ₅	4	2
P ₆	5	1

(7)

Pid	AT	BT
P ₁	2	3
P ₂	1	6
P ₃	2	4
P ₄	0	8
P ₅	1	2

(8)

Pid	AT	BT
P ₁	3	1
P ₂	1	4
P ₃	4	2
P ₄	0	6
P ₅	2	3

(9)

Pid	AT	BT
P ₁	2	6
P ₂	5	2
P ₃	1	8
P ₄	0	3
P ₅	4	4

(10)

Pid	AT	BT
P ₁	0	20
P ₂	15	25
P ₃	30	10
P ₄	45	15

(11)

Pid	AT	BT
P ₁	0	9
P ₂	1	4
P ₃	2	9

(12)

Pid	AT	BT
P ₁	0	7
P ₂	1	4
P ₃	2	8

③

Pid	AT	BT
P ₁	0	4
P ₂	1	2
P ₃	2	5
P ₄	3	4

④

Pid	AT	BT
P ₀	0	6
P ₁	1	3
P ₂	2	1
P ₃	3	4

⑤

Pid	AT	BT
P ₁	0	10
P ₂	0	1
P ₃	1	2
P ₄	2	4
P ₅	2	3

SJF - Pre-emptive Scheduling problems (SRTF)
[Shortest - Remaining time first]

Consider the set of processes whose arrival time & burst time are given below. If the CPU scheduling policy is SJF-Pre-emptive (SRTF). Calculate the average waiting time & average turnaround time.

①

Pid	AT	BT
P ₁	0	8
P ₂	1	4
P ₃	2	9
P ₄	3	5

②

Pid	AT	BT
P ₁	0	12
P ₂	2	4
P ₃	3	6
P ₄	8	5

③

Pid	AT	BT
P ₁	0	3
P ₂	1	1
P ₃	2	1
P ₄	3	2

Pid	AT	BT
P ₁	3	1
P ₂	1	4
P ₃	4	2
P ₄	0	6
P ₅	2	3

Pid	AT	BT
P ₁	2	6
P ₂	5	2
P ₃	1	8
P ₄	0	3
P ₅	4	4

Pid	AT	BT
P ₁	0	5
P ₂	8	4
P ₃	9	6
P ₄	8	2
P ₅	10	3

Pid	AT	BT
P ₁	2	3
P ₂	1	6
P ₃	2	4
P ₄	0	8
P ₅	1	2

Pid	AT	BT
P ₁	0	8
P ₂	1	4
P ₃	2	2
P ₄	3	1
P ₅	4	3
P ₆	5	2

Pid	AT	BT
P ₁	0	7
P ₂	1	5
P ₃	2	3
P ₄	3	1
P ₅	4	2
P ₆	5	1

Pid	AT	BT
P ₁	0	4
P ₂	1	2
P ₃	2	5
P ₄	3	4

Pid	AT	BT
P ₁	0	7
P ₂	3	2
P ₃	4	3
P ₄	4	1
P ₅	5	3

Pid	AT	BT
P ₁	0	20
P ₂	15	25
P ₃	30	10
P ₄	45	15

Pid	AT	BT
P ₁	0	10
P ₂	2	5
P ₃	3	2
P ₄	5	20

Pid	AT	BT
P ₁	0	8
P ₂	1	4
P ₃	2	9
P ₄	3	5

Processes	AT	BT
A	0	10
B	1	1
C	2	2
D	3	1
E	4	5

Priority Scheduling Problems:

Consider the following set of processes that arrive at the following time shown in table with burst time. Calculate Average waiting time & Average turnaround time by applying Priority Scheduling algorithm.

Non-Preemptive Priority Scheduling:

①

PID	BT	Priority	AT
P ₁	10	3	0
P ₂	1	1 (high)	0
P ₃	2	3	0
P ₄	1	4 (low)	0
P ₅	5	2	0

②

PID	AT	BT	Priority
P ₁	0	4	1
P ₂	0	3	2
P ₃	6	7	1 (h)
P ₄	11	4	3 (low)
P ₅	12	2	2

③

PID	AT	BT	Priority
P ₁	0	11	2
P ₂	5	28	0
P ₃	12	2	3
P ₄	9	10	1 (l)
P ₅	9	16	4 (h)

④

PID	AT	BT	Priority
P ₁	0	4	2 (l)
P ₂	1	3	3
P ₃	2	1	4
P ₄	3	5	5 (h)
P ₅	4	2	5

⑤

PID	AT	BT	Priority
P ₁	0	6	4
P ₂	3	5	2 (l)
P ₃	3	3	6 (h)
P ₄	5	5	3

⑥

PID	AT	BT	Priority
P ₁	0	10	2
P ₂	2	5	1
P ₃	3	2	0
P ₄	5	20	3

Pre-emptive Priority Scheduling:

①

PID	AT	BT	Priority
P ₁	0	7	3
P ₂	3	2	2
P ₃	4	3	1 (h)
P ₄	4	1	1
P ₅	5	3	3 (l)

②

PID	AT	BT	Priority
P ₁	0	4	2 (l)
P ₂	1	2	4
P ₃	2	3	6
P ₄	3	5	10
P ₅	4	1	8
P ₆	5	4	12 (h)
P ₇	6	6	9

③

PID	AT	BT	Priority
P ₁	2	4	3
P ₂	1	3	1 (h)
P ₃	0	2	4 (l)
P ₄	3	8	2

④

PID	AT	BT	Priority
P ₁	0	4	3
P ₂	1	2	2
P ₃	2	3	4 (low)
P ₄	4	2	1 (high)

⑤

PID	AT	BT	Priority
P ₁	0	11	2
P ₂	5	28	0 (high)
P ₃	12	2	3
P ₄	2	10	1
P ₅	9	16	4 (low)

⑥

PID	AT	BT	Priority
P ₁	0	8	3
P ₂	1	2	4
P ₃	3	4	4
P ₄	4	1	5
P ₅	5	6	2
P ₆	6	5	6 (low)
P ₇	10	1	1 (h)

⑦

PID	AT	BT	PT
P ₁	0	4	2 (high)
P ₂	1	3	3
P ₃	2	1	4
P ₄	3	5	5 (low)
P ₅	4	2	5

⑧

PID	AT	BT	Priority
P ₁	0	3	3
P ₂	1	4	2
P ₃	2	6	4
P ₄	3	4	6
P ₅	5	2	10

PID	AT	BT	Priority
P ₁	0	1	2 (low)
P ₂	1	7	6
P ₃	2	3	3
P ₄	3	6	5
P ₅	4	5	4
P ₆	5	15	10 (high)
P ₇	15	8	9

PID	AT	BT	Priority
P ₁	0	4	2
P ₂	1	3	1 (h)
P ₃	2	1	3
P ₄	3	5	4 (l)

Round Robin - Scheduling algorithm Problems:

①

PID	AT	BT
P ₁	0	10
P ₂	0	1
P ₃	0	2
P ₄	0	1
P ₅	0	5

Time Quantum
TQ = 1ms

②

PID	AT	BT
P ₆	0	10
P ₁	0	5
P ₂	1	2
P ₃	2	1

TQ = 1

③

PID	AT	BT
P ₁	0	5
P ₂	2	2
P ₃	6	8
P ₄	12	4

TQ = 2ms

④

PID	AT	BT
P ₁	0	8
P ₂	1	4
P ₃	2	9
P ₄	3	5

TQ = 2ms

⑤

PID	AT	BT
P ₁	0	24
P ₂	0	3
P ₃	0	3

TQ = 4ms

⑥

PID	AT	BT
P ₁	0	5
P ₂	1	3
P ₃	2	1
P ₄	3	2
P ₅	4	3

TQ = 3ms

PID	AT	BT
P ₁	0	10
P ₂	1	1
P ₃	2	2
P ₄	3	1
P ₅	4	5

TQ = 2ms

PID	BT
P ₁	10
P ₂	29
P ₃	3
P ₄	7
P ₅	12

Arrival time is
0 for all processes

TQ = 10ms

PID	AT	BT
P ₁	0	4
P ₂	1	5
P ₃	2	2
P ₄	3	1
P ₅	4	6
P ₆	6	3

TQ = 2ms

PID	AT	BT
P ₁	0	5
P ₂	1	6
P ₃	2	3
P ₄	3	1
P ₅	4	5
P ₆	6	4

TQ = 4ms

PID	AT	BT
P ₁	5	5
P ₂	4	6
P ₃	3	7
P ₄	1	9
P ₅	2	2
P ₆	6	3

TQ = 3ms

PID	AT	BT
P ₁	0	8
P ₂	1	6
P ₃	3	3
P ₄	5	2
P ₅	6	4

TQ = 4ms

PID	AT	BT
P ₁	0	4
P ₂	1	3
P ₃	2	7

TQ = 2ms

PID	AT	BT
P ₁	2	8
P ₂	0	7
P ₃	1	9

TQ = 4ms

PID	AT	BT
P ₁	0	4
P ₂	1	3
P ₃	2	1
P ₄	3	5
P ₅	4	2

TQ = 2ms